ADJUNCTION AND VANISHING FOR COHOMOLOGY WITH SUPPORT OF A SHEAF PUSHED FORWARD FROM A CLOSED SUBSET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites three cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- Derived adjunction from an adjoint pair of exact functors. Stacks Project, Derived Categories, Lemma derived-lemma-derived-adjoint-functors (tag 09T5). Transfers the abelian adjunction of sections with support to the derived categories.
- Pushforward of a K-injective complex is K-injective and the abelian adjunction of sections with support. Stacks Project, Cohomology of Sheaves, Lemma lemma-K-injective-flat (tag 0D5Z); Modules on Sites, Lemma modules-lemma-adjoint-section-with-support (tag 0G6P). The first computes R-H_Z(i_*K) from a K-injective representative and the second supplies the abelian adjunction.

Source reference. Original reference 0G6P: sources/stacks-chow/tags.txt, source lines 20204–20204. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 09T5: sources/stacks-chow/tags.txt, source lines 12378–12378. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0D5Z: sources/stacks-chow/tags.txt, source lines 16504–16504. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $i : Z \to X$ be the inclusion of a closed subset.*

(1) *$R\mathcal{H}_Z : D(\mathcal{O}_X) \to D(\mathcal{O}_X|_Z)$ is right adjoint to $i_* : D(\mathcal{O}_X|_Z) \to D(\mathcal{O}_X)$.*
(2) *For K in $D(\mathcal{O}_X|_Z)$ we have $R\mathcal{H}_Z(i_*K) = K$.*
(3) *Let $\mathcal{G}$ be a sheaf of $\mathcal{O}_X|_Z$-modules on Z. Then $\mathcal{H}^p_Z(i_*\mathcal{G}) = 0$ for $p > 0$.*

Proof. The functor i_* is exact, so $i_* = Ri_* = Li_*$. Hence part (1) of the lemma follows from Modules, Lemma 0G6P and Derived Categories, Lemma 09T5. Let K be as in (2). We can represent K by a K-injective complex $\mathcal{I}^\bullet$ of $\mathcal{O}_X|_Z$-modules. By Lemma 0D5Z the complex $i_*\mathcal{I}^\bullet$, which represents i_*K, is a K-injective complex of $\mathcal{O}_X$-modules. Thus $R\mathcal{H}_Z(i_*K)$ is computed by $\mathcal{H}_Z(i_*\mathcal{I}^\bullet) = \mathcal{I}^\bullet$ which proves (2). Part (3) is a special case of (2). □

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.